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1  #include <bits/stdc++.h>
2  using namespace std;
3  const int N=1e3+7,S=(1<<12);
4  long long n,a[4][N],dp[N][S],w[8][12],col[S<<3][4];
5  int getcol(int t,int b,int x,int y){//在一个大小为3*b的网格中，压缩值为t，询问(x,y)
    格子上的颜色
6  //红色为+1,黑色为0,蓝色为-1
7      memset(w,-1,sizeof(w));
8      for(int y=b;y>=1;y--){
9          for(int x=3;x>=1;x--){
10             w[x+2][y+2]=(t&1),t=t/2;//扩大一圈，避免越界
11         }
12     }
13     x+=2,y+=2;
14     int flag=0;
15     if(w[x][y]==w[x-1][y]&&w[x-1][y]==w[x-2][y]) flag=1;
16     if(w[x][y]==w[x-1][y]&&w[x-1][y]==w[x+1][y]) flag=1;
17     if(w[x][y]==w[x+1][y]&&w[x+1][y]==w[x+2][y]) flag=1;
18     if(w[x][y]==w[x][y-1]&&w[x][y]==w[x][y+1]) flag=1;
19     if(w[x][y]==w[x][y-1]&&w[x][y]==w[x][y-2]) flag=1;
20     if(w[x][y]==w[x][y+1]&&w[x][y]==w[x][y+2]) flag=1;
21     if(w[x][y]==w[x+1][y+1]&&w[x+1][y+1]==w[x+2][y+2]) flag=1;
22     if(w[x][y]==w[x-1][y-1]&&w[x-1][y-1]==w[x+1][y+1]) flag=1;
23     if(w[x][y]==w[x-1][y-1]&&w[x-1][y-1]==w[x-2][y-2]) flag=1;
24     if(w[x][y]==w[x-1][y+1]&&w[x][y]==w[x-2][y+2]) flag=1;
25     if(w[x][y]==w[x-1][y+1]&&w[x][y]==w[x+1][y-1]) flag=1;
26     if(w[x][y]==w[x+1][y-1]&&w[x][y]==w[x+2][y-2]) flag=1;
27     if(!flag) return 0;
28     if(!w[x][y]) return 1;
29     return -1;
30 }
31 int main(){
32     freopen("chess.in","r",stdin);
33     freopen("chess.out","w",stdout);
34     cin>>n;
35     for(int i=1;i<=3;i++){
36         for(int j=1;j<=n;j++){
37             cin>>a[i][j];
38         }
39         if(n<=5){//n<=5暴力
40             long long ans=-1e18;
41             for(int j=0;j<(1<<(3*n));j++){
42                 long long sum=0;
43                 for(int x=1;x<=3;x++){
44                     for(int y=1;y<=n;y++){
45                         sum+=getcol(j,n,x,y)*a[x][y];
46                     }
47                     ans=max(ans,sum);
48                 }
49             }
50             cout<<ans<<'\n'; return 0;
51         }
52         for(int j=0;j<(1<<15);j++){
53             for(int k=1;k<=3;k++){
54                 col[j][k]=getcol(j,5,k,3);
55             }
56         }
57         for(int i=4;i<=n;i++){
58             for(int j=0;j<(1<<12);j++) dp[i][j]=-1e18;
59         }
60     }
61 }

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54     for(int j=0;j<(1<<12);j++){
55         long long ans=0;
56         for(int x=1;x<=3;x++){
57             for(int y=1;y<=2;y++){
58                 ans+=getcol(j<<3,5,x,y)*a[x][y];
59             }
60             dp[4][j]=ans;//加上第一列第二列的答案
61         }
62         for(int i=4;i<=n;i++){
63             for(int j=0;j<(1<<12);j++){
64                 for(int k=0;k<8;k++){
65                     int t=((j&((1<<9)-1))<<3)+k;
66                     int s=(j<<3)+k;
67                     dp[i+1][t]=max(dp[i+1][t],col[s][1]*a[1][i-1]+col[s][2]*a[2]
[i-1]+col[s][3]*a[3][i-1]+dp[i][j]);
68                 }
69             }
70             long long ans=-1e18;
71             for(int j=0;j<(1<<12);j++){
72                 long long sum=dp[n][j];
73                 for(int x=1;x<=3;x++){
74                     for(int y=4;y<=5;y++){
75                         sum+=getcol(j,5,x,y)*a[x][n+y-5];//加上第n-1列第n列的答案
76                     }
77                     ans=max(ans,sum);
78                 }
79             cout<<ans<<'\n';
80             return 0;
81         }

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